Angularly Stable Ultrawide Band Single-Layer Frequency Selective Absorber

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***Abstract*- An ultrawide band angular stable frequency selective surface (FSS) absorber is proposed. The absorber consists of a single FSS lossy layer, a metal ground below and a F4BM350 dielectric layer on top. Lumped resistors are embedded at the joints of four trapezoidal loops to introduce resistance loss in the lossy layer, resulting in multimode resonance. To improve the absorption bandwidth, a dielectric matching layer is introduced. The wideband absorption mechanism is explored through an equivalent circuit model, input impedance and surface current distribution. The proposed absorber offers an over 90 % absorptivity bandwidth from 4.18 GHz to 22.39 GHz with a fractional bandwidth (FBW) of 137.1 %. The presented FSS unit cell has a surface area of $0.117\lambda_L \times 0.117\lambda_L$ and a thickness of $0.115\,\lambda_L$, where λ_L is the wavelength at 4.18 GHz. Simulation results demonstrate that the designed absorber achieves angular stability of 50° for both polarizations.**

***Index Terms*- Absorber, angular stability bandwidth, frequency selective surface,**

I. INTRODUCTION

Compared to conventional electromagnetic absorbers, frequency selective surface (FSS) absorbers offer an ultrawide bandwidth with relatively thinner thickness in the interested frequency range due to the flexible manipulation of electromagnetic waves by FSS. FSS absorbers have been extensively applied in various fields such as electromagnetic inference suppression and stealth technology. With the rapid development of wireless technology and radar technology, there is an increasing demand for ultrawide band, high angularly stable and thin absorbers. [1]

To reduce the overall thickness of the absorber, magnetic materials have been employed as substrate of FSS absorbers. For instance, T. Deng et al. proposed an absorber operating from 7.9 GHz to 14.6 GHz with a thickness of only 1.1 mm ($0.029\,\lambda_L$) [1]. However, magnetic materials are typically heavy and bulky, which may be impractical in applications where lightweight is crucial. Various techniques have been explored to enhance the incident angular stability of FSS absorbers, including miniaturization [2], 3-D structure [3], and use of dielectric compensation layers [4].

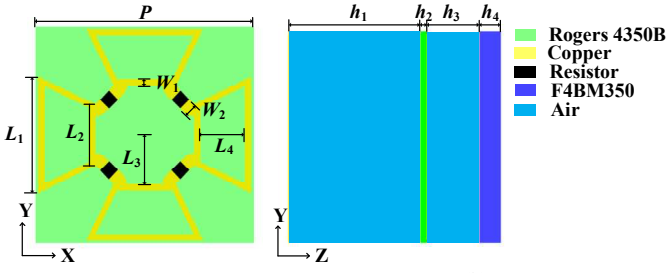
To expand the absorption bandwidth of FSS absorbers, several methods have been developed, such as multimode resonance [5], the combination of multiple resonance structures [6], multiple lossy FSS layers [7], 3-D structure [8], and the use of dielectric matching layers [9]. It is worthing noting that most wideband FSS absorbers are either multilayer FSS structures or single-layer configuration with multiple resonant structures. Multilayer FSS absorbers effectively avoid mutual coupling of different resonant structures and constraints on element dimensions. In contrast, single-layer absorbers incorporate multiple resonant structures in one layer, which poses a significant challenge in designing wideband and angularly stable FSS absorber. Notably, most existing wideband FSS absorbers integrate more than 4 resistors in a unit cell and have a fractional bandwidth below 130% [5][6].

In this paper, we propose a novel ultrawide band single-layer FSS absorber with high angular stability. Only 4 lumped resistors are loaded in the lossy FSS layer. The unit cell is designed to enhance the bandwidth by introducing multiple resonant modes and a 1 mm thick F4BM350 dielectric compensation layer. Simulation results demonstrate the proposed absorber offers an over 90% absorptivity fractional bandwidth of 137.1% with angular stability up to 50°.

II. DESIGN OF THE PROPOSED ABSORBER

A. Configuration of the FSS absorber

Fig. 1 illustrates the configuration of the proposed FSS absorber. The absorber is composed of five layers: a copper ground, a FSS lossy layer, a compensation dielectric layer, and two air layers to isolate them. The lossy FSS layer is printed on a 0.254 mm thick Rogers 4350B substrate with dielectric constant (ε_r) of 3.66 and loss tangent ($\tan\delta$) of 0.0037. The FSS unit cell consists of four trapezoidal loops, which are connected by four strip lines embedded with lumped resistors. The FSS lossy layer is separated from the ground by a 5 mm thick air layer. At a distance of 2 mm above the FSS layer, a 1 mm thick



(a) (b)

Figure 1. Configuration of the FSS unit cell (a) and side view (b) of the proposed absorber. P = 8.4, W_1 = 0.25, W_2 = 0.5, L_1 = 4.5, L_2 = 2.4, L_3 = 1.9, L_4 = 1.7, h_1 = 5, h_2 = 0.254, h_3 = 2, h_4 = 1 and R = 147 Ω
(All dimensions are in mm)

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F4BM350 slab with dielectric constant of 3.50 and loss tangent of 0.0012 is introduced to enhance both the bandwidth and angular stability. The values of all configuration parameters are listed in Fig. 1.

B. Design of the unit cell

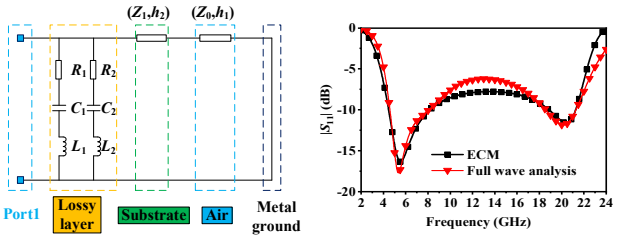
Based on transmission line (TL) theory, the equivalent circuit model (ECM) of the FSS absorber without the compensation slab is depicted in Fig. 2(a). The grounded air layer is equivalent to a shorted TL with length of h_1 and impedance of $Z_0 = 377\ \Omega$. The Rogers 4350B substrate is also considered as a transmission line with an impedance of $Z_1 = Z_0/\sqrt{\varepsilon_r}$, where ε_r is the dielectric constant of the substrate.

The FSS layer is regarded as a parallel connection of two different series RLC circuit. This is because the absorber exhibits two distinct resonance modes at low and high frequency, as shown in Fig. 3. It is evident that the intensity and direction of surface currents density differ significantly at 5.5 GHz and 20.5GHz corresponding to the two absorption peaks of the absorber. The equivalent inductance is larger at 5.5 GHz while smaller at 20.5 GHz. The reflection coefficient of the absorber (S_{11}) can then be given as

$$S_{11} = 20\lg|(Z_{in} - Z_0)/(Z_{in} + Z_0)| \quad (1)$$

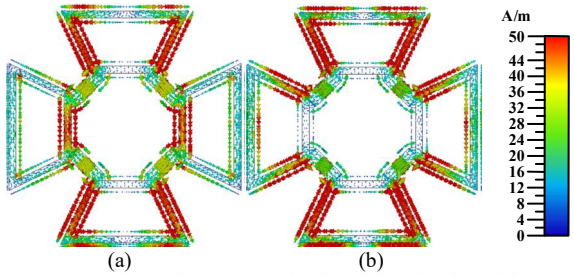
where Z_{in} is the input impedance of the absorber.

The reflection coefficient of the absorber without the compensation slab is analyzed using the ECM and compared to the results obtained from full wave analysis results (Fig. 2(b)). It is clear that the bandwidth for $|S_{11}| \le$ -10 dB is only around the two absorption peaks. Therefore, the F4BM350 compensation layer is introduced to improve the impedance matching.



(a) (b)

Figure 2. (a) ECM of the absorber without compensation slab, and (b) reflection coefficient of the absorber. $R_1 = 173\ \Omega$, $C_1 = 0.054$ pF, $L_1 = 3.586$ nH, $R_2 = 343\ \Omega$, $C_2 = 0.063$ pF, $L_2 = 9.786$ nH.



(a) (b)

Figure 3. The surface current density of the absorber without compensation slab at (a) 5.5 GHz, (b) 20.5GHz for TE polarization.

As illustrated in Fig. 4(a), the addition of the compensation slab enables the absorber to achieve an ultrawide absorption band with reflection below -10 dB in the range of 4.18 - 22.39 GHz. This attributes to: 1) the improved impedance matching of the absorber and 2) the introduction of a new absorption peak at the central frequency, about 13.8 GHz. By introducing the compensation slab, the normalized reactance of the absorber approaches to zero in a wide frequency range, while the normalized resistance fluctuates around 1, as shown in Fig. 4(b). This implies excellent impedance matching to impedance in free space in the frequency range of 4.18 - 22.39 GHz.

The thickness of the compensation slab significantly impacts the bandwidth, as indicated in Fig. 5. The frequency of the new absorption peak decreases as the thickness of the slab increase, while the impedance matching degrades at higher frequency bands. Finally, a thickness of 1 mm is determined, taking both bandwidth and commercial availability into consideration.

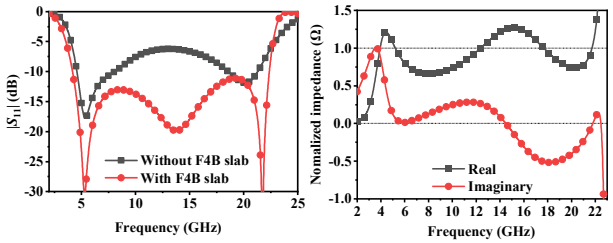
III. PERFORMANCE ANALYSIS

The absorptivity (A) of the absorber can be calculated using the following formula:

$$A = 1 - |S_{11}|^2 - |S_{21}|^2 \quad (2)$$

where $|S_{21}|$ is the transmission coefficient of the absorber. Since a copper ground is employed in the absorber, $|S_{21}|$ equals to 0, and thus the absorptivity A is solely determined by $|S_{11}|$. As a results, values of $|S_{11}| \le$ - 10 dB correspond to $A \ge 90\%$.

The polarization insensitivity of the proposed absorber is illustrated in Fig. 6. The coincidence of three absorptivity curves for different polarization angles demonstrates that the proposed



(a) (b)

Figure 4. (a) $|S_{11}|$ of the absorber, and (b) the normalized impedance (Z_{in} / 377 Ω) of the absorber with the compensation slab.

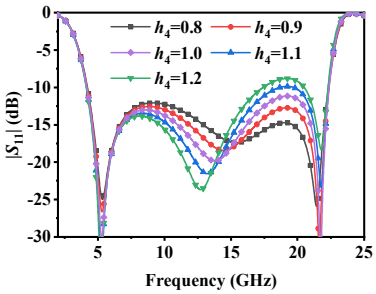


Figure 5. The impact of compensation slab thicknesses (in mm) on $|S_{11}|$ of the absorber.

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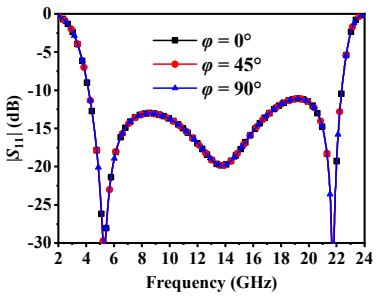


Figure 6. Polarization insensitivity of the proposed FSS absorber.

absorber is independent on the polarization of incident electromagnetic waves.

Angular stability is one of important indexes of electromagnetic absorbers. Therefore, the angular stability of the proposed FSS absorber is examined, as depicted in Fig. 7. It can be observed that the absorber maintains an absorptivity over 90% in the frequency range from 4.18 - 22.39 GHz with a fractional bandwidth (FBW) of 137.1% for normal incidence. For oblique incidence, the absorber achieves an FBW of 102.5% (6.50- 20.17GHz) with absorptivity over 80% when the incident angle increases up to 50° for both TE and TM polarizations.

Table I presents a comparison of the performance of the proposed absorber with some resistors-embedded absorbers from literature. The propose absorber offers a wider bandwidth, good angular stability and a better trade-off between the thickness and bandwidth while utilizing only 4 resistors per unit cell.

IV. CONCLUSION

This paper presents a single-layer frequency selective absorber that is ultrawide band and angularly stable, with only 4 resistors embedded per unit cell. To improve the bandwidth of the absorber, both multimode resonance and a dielectric compensation layer are considered. The proposed absorber achieves absorptivity over 90% in the frequency range from 4.18 to 22.39 GHz with a fractional bandwidth of 137.1%. For oblique incidence up to 50°, the absorptivity remains above 80% in the frequency range of 6.50 - 20.17 GHz for both TE and TM polarizations. With its properties, the proposed FSS absorber shows potential applications in electromagnetic interference suppression and design of absorption/transmission structures.

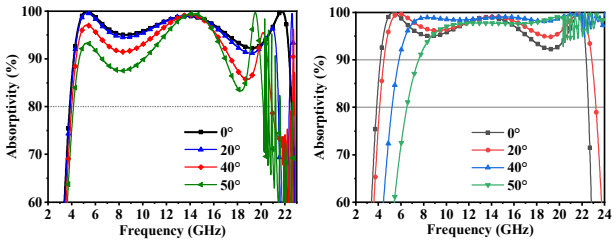


Figure 7. Angular stability of the proposed FSS absorber for (a) TE and (b) TM polarizations.

TABLE I
PERFORMANCE COMPARISON

Ref.	Bandwidth (GHz)	Thickness	AS (°) (TE, TM)	FSS layers	RPU
[6]	6.7 - 20.58 (101.7%)	$0.067\lambda_L$	30, 45	1	8
[7]	2.24 - 11.4 (134.3%)	$0.075\lambda_L$	45,45	2	16
[10]	2.7 - 12.7 (130. 0%)	$0.084\lambda_L$	40, 40	1	8
[11]	3.87 - 14.84 (117.0%)	$0.19\lambda_L$	40, 50	2	8
[12]	1.03 - 13.27 (171.2%)	$0.101\lambda_L$	-, -	2	6
[13]	5.44 - 10.21 (61.2%)	$0.29\lambda_L$	50 - 80	1	4
This work	4.18 - 22.39 (137.1%)	$0.115\lambda_L$	50, 50	1	4

AS: Angular stability with absorptivity larger than 80%.

RPU: Resistors loaded per unit

REFERENCES

[1] T. Deng, Z.-W. Li, M.-J. Chua, and Z. N. Chen, "Broadband and ultrathin frequency-dispersive metamaterial screen for reflectivity reduction," *IEEE Trans. Antennas Propag.*, vol. 63, no. 9, pp. 4156–4160, Sep. 2015.

[2] Y.-T. Zhao, B. Chen, and B. Wu, "Miniaturized periodicity broadband absorber with via-based hybrid metal-graphene structure for large-angle RCS reduction," *IEEE Trans. Antennas Propag.*, vol. 70, no. 4, pp. 2832–2840, Apr. 2022.

[3] T. Shi, L. Jin, L. Han, M. Tang et al., "Dispersion-engineered, broadband, wide-angle, polarization-independent microwave metamaterial absorber," *IEEE Trans. Antennas Propag.*, vol. 69, no. 1, pp. 229–238, Jan. 2021.

[4] Z. Yao, S. Xiao, Y. Li, and B.-Z. Wang, "Wide-angle, ultra-wideband, polarization-independent circuit analog absorbers," *IEEE Trans. Antennas Propag.*, vol. 70, no. 8, pp. 7276–7281, Aug. 2022.

[5] B. Zhang, C. Jin, and Z. Shen, "Low-profile broadband absorber based on multimode resistor-embedded metallic strips," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 3, pp. 835–843, Mar. 2020.

[6] S. Sambhav, J. Ghosh, and A. K. Singh, "Ultra-wideband polarization insensitive thin absorber based on resistive concentric circular rings," *IEEE Trans. Electromagn. Compat.*, vol. 63, no. 5, pp. 1333–1340, Oct. 2021.

[7] Z. Yao, S. Xiao, Z. Jiang, L. Yan, and B.-Z. Wang, "On the design of ultrawideband circuit analog absorber based on quasi-single-layer fss," *IEEE Antennas Wireless Propag. Lett.*, vol. 19, no. 4, pp. 591–595, Apr. 2020.

[8] G. Luo, W. Yu, Y. Yu, X. Zhang, and Z. Shen, "A three-dimensional design of ultra-wideband microwave absorbers," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 10, pp. 4206–4215, Oct. 2020.

[9] Z. Ma, C. Jiang, W. Cao, J. Li, and X. Huang, "An ultrawideband and high-absorption circuit-analog absorber with incident angle-insensitive performance," *IEEE Trans. Antennas Propag.*, vol. 70, no. 10, pp. 9376–9384, Oct. 2022.

[10] Y. Zhang, W. Yang, X. Li, and G. Liu, "Design and analysis of a broadband microwave metamaterial absorber," *IEEE Photonics J.*, vol. 15, no. 3, pp. 1–10, Jun. 2023.

[11] D. Lim and S. Lim, "Ultrawideband electromagnetic absorber using sandwiched broadband metasurfaces," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, no. 9, pp. 1887–1891, Sep. 2019.

[12] J. Yu, Y. Cai, X. Lin, and X. Wang, "Perforated multilayer ultrawideband absorber based on circuit analog absorber with optimal air spaces," *IEEE Antennas Wireless Propag. Lett.*, vol. 19, no. 1, pp. 34–38, Jan. 2020.

[13] Q. Lv, C. Jin, B. Zhang, P. Zhang, J. Wang, et al., "Wideband dual-polarized microwave absorber at extremely oblique incidence," *IEEE Trans. Antennas Propag.*, vol. 71, no. 3, pp. 2497–2506, Mar. 2023.

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